

# SUPERMART MANAGEMENT SYSTEM

## API Reference Documentation

---

Version: v1.0 | Base URL: <https://api.supermart.vn/api/v1>

Project Code: SMMS | February 2026

### HTTP Method Legend

|     |      |     |       |        |
|-----|------|-----|-------|--------|
| GET | POST | PUT | PATCH | DELETE |
|-----|------|-----|-------|--------|

All requests require **Authorization: Bearer <token>** header unless noted otherwise.

All responses use **Content-Type: application/json** format.

# 1. Identity Service

## 1.1 Authentication & User Management APIs

Manages user authentication, JWT token issuance, password management, and user account CRUD operations. Enforces Role-Based Access Control (RBAC) across all services.

**POST** `/api/v1/auth/login`

### User Login

Authenticate user credentials and return a JWT access token with refresh token.

 **Auth:** None (public endpoint)

### Request Body

| Field    | Type   | Required | Constraints  | Description                                   |
|----------|--------|----------|--------------|-----------------------------------------------|
| username | string | Yes      | Max 50 chars | Account username                              |
| password | string | Yes      | Min 8 chars  | Account password (plaintext, sent over HTTPS) |

Example:

```
{  "username": "cashier_alex",  "password": "P@ssw0rd123" }
```

### Response

| Field         | Type   | Description                              |
|---------------|--------|------------------------------------------|
| accessToken   | string | JWT access token (expires in 8 hours)    |
| refreshToken  | string | JWT refresh token (expires in 7 days)    |
| tokenType     | string | Always 'Bearer'                          |
| user.id       | UUID   | Internal user ID                         |
| user.username | string | Username                                 |
| user.role     | string | ROLE_ADMIN   ROLE_MANAGER   ROLE_CASHIER |
| user.status   | string | ACTIVE   LOCKED                          |

Example:

```
{  "accessToken": "eyJhbGc...",  "refreshToken": "eyJhbGc...",  "tokenType": "Bearer",  "user": {    "id": "uuid-...",    "username": "cashier_alex",    "role": "ROLE_CASHIER",    "status": "ACTIVE"  } }
```

### Error Responses

| HTTP Code | Error Code          | Description                                     |
|-----------|---------------------|-------------------------------------------------|
| 401       | INVALID_CREDENTIALS | Username or password is incorrect               |
| 403       | ACCOUNT_LOCKED      | User account has been locked by Admin           |
| 429       | TOO_MANY_ATTEMPTS   | Login attempts exceeded (5 times in 15 minutes) |

**POST** /api/v1/auth/refresh**Refresh Access Token**

Exchange a valid refresh token for a new access token.

 **Auth:** None

**Request Body**

| Field        | Type   | Required | Constraints | Description                       |
|--------------|--------|----------|-------------|-----------------------------------|
| refreshToken | string | Yes      | Valid JWT   | Refresh token obtained from login |

**Example:**

```
{ "refreshToken": "eyJhbGc..." }
```

**Response**

| Field        | Type   | Description                 |
|--------------|--------|-----------------------------|
| accessToken  | string | New JWT access token        |
| refreshToken | string | New refresh token (rotated) |

**Error Responses**

| HTTP Code | Error Code    | Description                         |
|-----------|---------------|-------------------------------------|
| 401       | INVALID_TOKEN | Refresh token is invalid or expired |

**POST** /api/v1/auth/logout**User Logout**

Invalidate the user's session by blacklisting the token.

 **Auth:** Bearer token (any role)

**Response****Example:**

```
{ "message": "Logged out successfully" }
```

**Error Responses**

| HTTP Code | Error Code   | Description             |
|-----------|--------------|-------------------------|
| 401       | UNAUTHORIZED | No valid token provided |

**POST** /api/v1/auth/forgot-password**Forgot Password - Request OTP**

Trigger an OTP to be sent to the user's registered email via SMS Provider.

 **Auth:** None

**Request Body**

| Field | Type   | Required | Constraints        | Description                             |
|-------|--------|----------|--------------------|-----------------------------------------|
| email | string | Yes      | Valid email format | Registered email address of the account |

**Example:**

```
{ "email": "john.doe@supermart.vn" }
```

## Response

**Example:**

```
{ "message": "OTP sent to email", "expiredAt": "2026-02-01T10:15:00Z" }
```

## Error Responses

| HTTP Code | Error Code      | Description                           |
|-----------|-----------------|---------------------------------------|
| 404       | EMAIL_NOT_FOUND | No account associated with this email |
| 429       | RATE_LIMIT      | OTP request limit reached             |

## POST /api/v1/auth/reset-password

### Reset Password with OTP

Verify OTP and set a new password for the account.

 **Auth:** None

#### Request Body

| Field       | Type   | Required | Constraints | Description                |
|-------------|--------|----------|-------------|----------------------------|
| email       | string | Yes      | -           | Email used to request OTP  |
| otp         | string | Yes      | 6 digits    | OTP received via email/SMS |
| newPassword | string | Yes      | Min 8 chars | New password to set        |

## Error Responses

| HTTP Code | Error Code  | Description                 |
|-----------|-------------|-----------------------------|
| 400       | INVALID_OTP | OTP is incorrect or expired |

## GET /api/v1/users

### Get All User Accounts

Return paginated list of all staff accounts. Admin only.

 **Auth:** Bearer token — ROLE\_ADMIN

#### Query Parameters

| Parameter | Type    | Required | Default | Description |
|-----------|---------|----------|---------|-------------|
| page      | integer | No       | 1       | Page number |

| Parameter | Type    | Required | Default   | Description                                       |
|-----------|---------|----------|-----------|---------------------------------------------------|
| size      | integer | No       | 20        | Records per page (max 100)                        |
| role      | string  | No       | -         | Filter by role:<br>ROLE_MANAGER  <br>ROLE_CASHIER |
| status    | string  | No       | -         | Filter by status: ACTIVE  <br>LOCKED              |
| search    | string  | No       | -         | Search by username or email<br>(partial match)    |
| sortBy    | string  | No       | createdAt | Sort field: createdAt   lastLogin<br>  username   |

## Response

| Field               | Type     | Description                |
|---------------------|----------|----------------------------|
| content             | array    | List of user objects       |
| content[].id        | UUID     | User ID                    |
| content[].username  | string   | Username                   |
| content[].email     | string   | Email address              |
| content[].role      | string   | Assigned role              |
| content[].status    | string   | ACTIVE   LOCKED            |
| content[].lastLogin | datetime | Last login timestamp       |
| content[].createdAt | datetime | Account creation timestamp |
| totalElements       | integer  | Total number of records    |
| totalPages          | integer  | Total pages                |

## Error Responses

| HTTP Code | Error Code | Description                   |
|-----------|------------|-------------------------------|
| 403       | FORBIDDEN  | Insufficient role permissions |

**POST** /api/v1/users

## Create New User Account

Create a new staff account (Manager or Cashier). System sends credentials to their email.

 **Auth:** Bearer token — ROLE\_ADMIN

## Request Body

| Field    | Type   | Required | Constraints    | Description    |
|----------|--------|----------|----------------|----------------|
| username | string | Yes      | Unique, Max 50 | Login username |

| Field    | Type    | Required | Constraints               | Description                           |
|----------|---------|----------|---------------------------|---------------------------------------|
| email    | string  | Yes      | Unique, valid format      | Email for notifications               |
| roleId   | integer | Yes      | Must exist in roles table | Role to assign (Manager=1, Cashier=2) |
| fullName | string  | Yes      | Max 100                   | Display name                          |

**Example:**

```
{  "username": "manager_lan",    "email": "lan@supermart.vn",    "roleId": 1,    "fullName": "Nguyen Thi Lan" }
```

**Response**

| Field     | Type     | Description               |
|-----------|----------|---------------------------|
| id        | UUID     | New user ID               |
| username  | string   | Created username          |
| email     | string   | Email address             |
| role      | string   | Assigned role name        |
| status    | string   | Always ACTIVE on creation |
| createdAt | datetime | Creation timestamp        |

**Error Responses**

| HTTP Code | Error Code        | Description              |
|-----------|-------------------|--------------------------|
| 409       | USERNAME_CONFLICT | Username already exists  |
| 409       | EMAIL_CONFLICT    | Email already registered |

**GET** /api/v1/users/{id}

**Get User Account Detail**

Retrieve full details of a specific user account.

 **Auth:** Bearer token — ROLE\_ADMIN

**Path Parameters**

| Parameter | Type | Required | Description              |
|-----------|------|----------|--------------------------|
| id        | UUID | Yes      | User's unique identifier |

**Response**

| Field    | Type   | Description |
|----------|--------|-------------|
| id       | UUID   | User ID     |
| username | string | Username    |

| Field     | Type     | Description              |
|-----------|----------|--------------------------|
| email     | string   | Email                    |
| role      | object   | Role object { id, name } |
| status    | string   | ACTIVE   LOCKED          |
| createdAt | datetime | Created time             |
| lastLogin | datetime | Last login time          |

### Error Responses

| HTTP Code | Error Code     | Description                 |
|-----------|----------------|-----------------------------|
| 404       | USER_NOT_FOUND | No user found with given ID |

**PUT** /api/v1/users/{id}

### Update User Account

Update user email, role, or full name.

 **Auth:** Bearer token — ROLE\_ADMIN

### Path Parameters

| Parameter | Type | Required | Description       |
|-----------|------|----------|-------------------|
| id        | UUID | Yes      | User ID to update |

### Request Body

| Field    | Type    | Required | Constraints   | Description         |
|----------|---------|----------|---------------|---------------------|
| email    | string  | No       | Valid email   | New email address   |
| roleId   | integer | No       | Valid role ID | New role assignment |
| fullName | string  | No       | Max 100       | Updated full name   |

### Error Responses

| HTTP Code | Error Code     | Description        |
|-----------|----------------|--------------------|
| 404       | USER_NOT_FOUND | User not found     |
| 409       | EMAIL_CONFLICT | Email already used |

**PATCH** /api/v1/users/{id}/status

### Lock / Unlock User Account

Toggle user account status between ACTIVE and LOCKED. Locked users cannot login. Action is logged in audit trail.

 **Auth:** Bearer token — ROLE\_ADMIN

### Path Parameters

| Parameter | Type | Required | Description |
|-----------|------|----------|-------------|
| id        | UUID | Yes      | User ID     |

### Request Body

| Field  | Type   | Required | Constraints     | Description                      |
|--------|--------|----------|-----------------|----------------------------------|
| status | string | Yes      | ACTIVE   LOCKED | New account status               |
| reason | string | No       | Max 200         | Reason for locking (recommended) |

### Error Responses

| HTTP Code | Error Code     | Description                     |
|-----------|----------------|---------------------------------|
| 400       | INVALID_STATUS | Status must be ACTIVE or LOCKED |

**POST** /api/v1/users/{id}/reset-password

### Admin Reset User Password

Admin force-resets a staff member's password. New temporary password is sent to user's email.

 **Auth:** Bearer token — ROLE\_ADMIN

### Path Parameters

| Parameter | Type | Required | Description |
|-----------|------|----------|-------------|
| id        | UUID | Yes      | User ID     |

### Response

Example:

```
{ "message": "Password reset email sent to user's registered email" }
```

### Error Responses

| HTTP Code | Error Code     | Description    |
|-----------|----------------|----------------|
| 404       | USER_NOT_FOUND | User not found |

**GET** /api/v1/roles

### Get All Roles

Retrieve all available roles and their permissions.

 **Auth:** Bearer token — ROLE\_ADMIN

### Response

| Field          | Type    | Description                                                    |
|----------------|---------|----------------------------------------------------------------|
| [].id          | integer | Role ID                                                        |
| [].name        | string  | Role name (ROLE_ADMIN, ROLE_MANAGER, ROLE_CASHIER)             |
| [].permissions | array   | List of permission objects { id, permissionCode, description } |

## 2. Staff Service

### 2.1 Human Resource Management APIs

Handles employee HR profiles, shift definitions and scheduling, attendance tracking (check-in/check-out), leave request workflows, and automated payroll calculation.

**GET** `/api/v1/employees`

#### Get All Employees

Return paginated list of all employees with HR profile information.

 **Auth:** Bearer token — ROLE\_ADMIN | ROLE\_MANAGER

#### Query Parameters

| Parameter    | Type    | Required | Default | Description                     |
|--------------|---------|----------|---------|---------------------------------|
| page         | integer | No       | 1       | Page number                     |
| size         | integer | No       | 20      | Records per page                |
| departmentId | integer | No       | -       | Filter by department            |
| search       | string  | No       | -       | Search by name, phone, or email |

#### Response

| Field                | Type    | Description                |
|----------------------|---------|----------------------------|
| content[].id         | UUID    | Employee ID                |
| content[].fullName   | string  | Full name                  |
| content[].phone      | string  | Contact phone              |
| content[].email      | string  | Work email                 |
| content[].baseSalary | decimal | Base salary (VND)          |
| content[].department | object  | { id, name }               |
| content[].accountId  | UUID    | Linked identity account ID |

**POST** `/api/v1/employees`

#### Create Employee Profile

Create a new employee HR profile linked to an existing identity account.

 **Auth:** Bearer token — ROLE\_ADMIN | ROLE\_MANAGER

#### Request Body

| Field     | Type   | Required | Constraints                    | Description              |
|-----------|--------|----------|--------------------------------|--------------------------|
| accountId | UUID   | Yes      | Must exist in Identity Service | Link to identity account |
| fullName  | string | Yes      | Max 255                        | Full name                |

| Field             | Type    | Required | Constraints | Description                     |
|-------------------|---------|----------|-------------|---------------------------------|
| phone             | string  | No       | Max 20      | Contact phone number            |
| email             | string  | No       | Valid email | Personal or work email          |
| address           | string  | No       | Max 500     | Residential address             |
| taxCode           | string  | No       | Max 50      | Personal tax code               |
| bankAccountNumber | string  | No       | Max 50      | Bank account number for payroll |
| baseSalary        | decimal | Yes      | > 0         | Agreed base salary (VND)        |
| departmentId      | integer | No       | Must exist  | Department to assign            |

### Error Responses

| HTTP Code | Error Code             | Description                                                 |
|-----------|------------------------|-------------------------------------------------------------|
| 409       | ACCOUNT_ALREADY_LINKED | The accountId is already linked to another employee profile |

**GET** `/api/v1/employees/{id}`

### Get Employee Detail

Retrieve full HR profile of a specific employee.

 **Auth:** Bearer token — ROLE\_ADMIN | ROLE\_MANAGER

### Path Parameters

| Parameter | Type | Required | Description |
|-----------|------|----------|-------------|
| id        | UUID | Yes      | Employee ID |

### Response

| Field             | Type    | Description               |
|-------------------|---------|---------------------------|
| id                | UUID    | Employee ID               |
| fullName          | string  | Full name                 |
| phone             | string  | Phone number              |
| email             | string  | Email                     |
| address           | string  | Address                   |
| taxCode           | string  | Tax code                  |
| bankAccountNumber | string  | Bank account              |
| baseSalary        | decimal | Base salary               |
| department        | object  | { id, name, description } |

| Field        | Type   | Description                 |
|--------------|--------|-----------------------------|
| currentShift | object | Current assigned shift info |

### PUT /api/v1/employees/{id}

#### Update Employee Profile

Update HR information of an existing employee.

 **Auth:** Bearer token — ROLE\_ADMIN | ROLE\_MANAGER

#### Path Parameters

| Parameter | Type | Required | Description |
|-----------|------|----------|-------------|
| id        | UUID | Yes      | Employee ID |

#### Request Body

| Field             | Type    | Required | Constraints | Description       |
|-------------------|---------|----------|-------------|-------------------|
| fullName          | string  | No       | Max 255     | Updated full name |
| phone             | string  | No       | Max 20      | Phone number      |
| address           | string  | No       | -           | Address           |
| baseSalary        | decimal | No       | > 0         | Base salary       |
| bankAccountNumber | string  | No       | -           | Bank account      |

### GET /api/v1/shifts

#### Get All Shift Definitions

Retrieve all defined work shifts (Morning, Afternoon, Evening).

 **Auth:** Bearer token — ROLE\_MANAGER

#### Response

| Field          | Type    | Description                                     |
|----------------|---------|-------------------------------------------------|
| [].id          | integer | Shift ID                                        |
| [].shiftName   | string  | Shift name (e.g., Morning)                      |
| [].startTime   | time    | Shift start time (HH:mm)                        |
| [].endTime     | time    | Shift end time (HH:mm)                          |
| [].coefficient | float   | Salary multiplier (e.g., 1.0, 1.5 for overtime) |

### GET /api/v1/shift-schedules

#### Get Work Schedule

Retrieve shift schedule for a given period, optionally filtered by employee.

 **Auth:** Bearer token — ROLE\_MANAGER | ROLE\_CASHIER

## Query Parameters

| Parameter  | Type | Required | Default            | Description                                         |
|------------|------|----------|--------------------|-----------------------------------------------------|
| employeeId | UUID | No       | -                  | Filter by employee (Cashier sees own schedule only) |
| from       | date | No       | Current week start | Start date (YYYY-MM-DD)                             |
| to         | date | No       | Current week end   | End date (YYYY-MM-DD)                               |

## Response

| Field       | Type    | Description                           |
|-------------|---------|---------------------------------------|
| [].id       | integer | Schedule ID                           |
| [].employee | object  | { id, fullName }                      |
| [].shift    | object  | { id, shiftName, startTime, endTime } |
| [].workDate | date    | Assigned work date                    |

## POST /api/v1/shift-schedules

### Assign Shift to Employee

Assign a specific shift to an employee on a given date.

 **Auth:** Bearer token — ROLE\_MANAGER

### Request Body

| Field      | Type    | Required | Constraints                 | Description         |
|------------|---------|----------|-----------------------------|---------------------|
| employeeId | UUID    | Yes      | Must exist                  | Employee to assign  |
| shiftId    | integer | Yes      | Must exist                  | Shift definition ID |
| workDate   | date    | Yes      | YYYY-MM-DD, future or today | Date of the shift   |

## Error Responses

| HTTP Code | Error Code     | Description                                        |
|-----------|----------------|----------------------------------------------------|
| 409       | SHIFT_CONFLICT | Employee already has a shift assigned on this date |

## POST /api/v1/attendance

### Record Attendance (Check-in / Check-out)

Record actual check-in or check-out time for an employee's shift.

 **Auth:** Bearer token — ROLE\_CASHIER | ROLE\_MANAGER

### Request Body

| Field      | Type | Required | Constraints | Description |
|------------|------|----------|-------------|-------------|
| employeeId | UUID | Yes      | -           | Employee ID |

| Field           | Type     | Required | Constraints          | Description                                   |
|-----------------|----------|----------|----------------------|-----------------------------------------------|
| shiftScheduleId | integer  | Yes      | Must exist           | The scheduled shift being recorded            |
| type            | string   | Yes      | CHECK_IN   CHECK_OUT | Type of attendance event                      |
| timestamp       | datetime | No       | ISO 8601             | Actual time (defaults to current server time) |

## Response

| Field        | Type     | Description                                        |
|--------------|----------|----------------------------------------------------|
| id           | integer  | Attendance log ID                                  |
| checkInTime  | datetime | Recorded check-in time                             |
| checkOutTime | datetime | Recorded check-out time (null if not yet)          |
| totalHours   | float    | Calculated hours worked (null if shift still open) |

**GET** /api/v1/leave-requests

## Get Leave Requests

Retrieve list of leave requests, filterable by status and employee.

 **Auth:** Bearer token — ROLE\_MANAGER | ROLE\_CASHIER

## Query Parameters

| Parameter  | Type   | Required | Default | Description                                    |
|------------|--------|----------|---------|------------------------------------------------|
| employeeId | UUID   | No       | -       | Filter by employee (Cashier sees own requests) |
| status     | string | No       | -       | PENDING   APPROVED   REJECTED                  |

## Response

| Field        | Type     | Description                   |
|--------------|----------|-------------------------------|
| [].id        | integer  | Request ID                    |
| [].employee  | object   | { id, fullName }              |
| [].startDate | datetime | Leave start date              |
| [].endDate   | datetime | Leave end date                |
| [].reason    | string   | Reason for leave              |
| [].status    | string   | PENDING   APPROVED   REJECTED |

**POST** /api/v1/leave-requests

## Submit Leave Request

Employee submits a new leave request for manager approval.

 **Auth:** Bearer token — ROLE\_CASHIER | ROLE\_MANAGER

### Request Body

| Field      | Type     | Required | Constraints         | Description                 |
|------------|----------|----------|---------------------|-----------------------------|
| employeeId | UUID     | Yes      | Own ID for cashiers | Employee submitting request |
| startDate  | datetime | Yes      | Future date         | Start of leave period       |
| endDate    | datetime | Yes      | After startDate     | End of leave period         |
| reason     | string   | Yes      | Max 500             | Reason for leave            |

**PATCH** /api/v1/leave-requests/{id}/status

## Approve / Reject Leave Request

Manager approves or rejects a pending leave request.

 **Auth:** Bearer token — ROLE\_MANAGER

### Path Parameters

| Parameter | Type    | Required | Description      |
|-----------|---------|----------|------------------|
| id        | integer | Yes      | Leave request ID |

### Request Body

| Field  | Type   | Required | Constraints         | Description                    |
|--------|--------|----------|---------------------|--------------------------------|
| status | string | Yes      | APPROVED   REJECTED | Decision                       |
| note   | string | No       | Max 300             | Optional note for the decision |

## Error Responses

| HTTP Code | Error Code        | Description                                  |
|-----------|-------------------|----------------------------------------------|
| 400       | ALREADY_PROCESSED | Leave request is no longer in PENDING status |

**GET** /api/v1/payrolls

## Get Payroll Records

Retrieve payroll records for a specific month. Calculated from attendance logs.

 **Auth:** Bearer token — ROLE\_MANAGER

### Query Parameters

| Parameter | Type    | Required | Default | Description  |
|-----------|---------|----------|---------|--------------|
| month     | integer | Yes      | -       | Month (1-12) |

| Parameter  | Type    | Required | Default | Description                  |
|------------|---------|----------|---------|------------------------------|
| year       | integer | Yes      | -       | Year (e.g., 2026)            |
| employeeId | UUID    | No       | -       | Filter for specific employee |

Response

| Field             | Type    | Description                 |
|-------------------|---------|-----------------------------|
| [].id             | integer | Payroll record ID           |
| [].employee       | object  | { id, fullName }            |
| [].month          | integer | Payroll month               |
| [].year           | integer | Payroll year                |
| [].totalWorkHours | float   | Total hours worked          |
| [].standardSalary | decimal | Calculated base pay         |
| [].bonus          | decimal | Additional bonuses          |
| [].deduction      | decimal | Deductions (tax, insurance) |
| [].finalSalary    | decimal | Net take-home salary        |
| [].status         | string  | DRAFT   PAID                |

**POST** /api/v1/payrolls/generate

Generate Payroll for Month

Batch job trigger: calculate and generate payroll records for all employees in a given month.

 **Auth:** Bearer token — ROLE\_MANAGER

Request Body

| Field | Type    | Required | Constraints | Description  |
|-------|---------|----------|-------------|--------------|
| month | integer | Yes      | 1-12        | Target month |
| year  | integer | Yes      | >= 2020     | Target year  |

Response

Example:

```
{  "message": "Payroll generated for 24 employees",  "month": 2,  "year": 2026,  "totalProcessed": 24 }
```

## 3. Product Service

### 3.1 Product Catalog APIs

Manages the complete product catalog including categories, product CRUD, search (via ElasticSearch), and image management. Publishes events to Kafka on product changes.

**GET** `/api/v1/products`

#### Search / List Products

Full-text search and filtered listing of all products. Used by both POS and management UI. Returns high-speed search results from ElasticSearch index.

 **Auth:** Bearer token — All roles

#### Query Parameters

| Parameter  | Type    | Required | Default | Description                                 |
|------------|---------|----------|---------|---------------------------------------------|
| q          | string  | No       | -       | Full-text search query (name, SKU, barcode) |
| categoryId | UUID    | No       | -       | Filter by category                          |
| isActive   | boolean | No       | true    | Filter by active status                     |
| page       | integer | No       | 1       | Page number                                 |
| size       | integer | No       | 20      | Records per page                            |
| sortBy     | string  | No       | name    | Sort field: name   price   createdAt        |

#### Response

| Field               | Type    | Description                       |
|---------------------|---------|-----------------------------------|
| content[].id        | UUID    | Product ID                        |
| content[].sku       | string  | Stock Keeping Unit / Barcode      |
| content[].name      | string  | Product display name              |
| content[].price     | decimal | Retail price (VND)                |
| content[].unit      | string  | Unit of measure (e.g., Piece, Kg) |
| content[].isActive  | boolean | Visibility status                 |
| content[].category  | object  | { id, name }                      |
| content[].imageUrls | array   | List of product image URLs        |
| totalElements       | integer | Total count                       |

**POST** `/api/v1/products`

#### Create New Product

Create a new product in the catalog. Upon success, publishes ProductCreated event to Kafka to notify Inventory Service.

 **Auth:** Bearer token — ROLE\_MANAGER

**Request Body**

| Field       | Type          | Required | Constraints    | Description                                              |
|-------------|---------------|----------|----------------|----------------------------------------------------------|
| sku         | string        | Yes      | Unique, Max 50 | Barcode / Stock Keeping Unit                             |
| name        | string        | Yes      | Max 255        | Product display name                                     |
| price       | decimal       | Yes      | > 0            | Retail selling price (VND)                               |
| unit        | string        | Yes      | Max 20         | Unit of measure: Piece, Kg, Liter, Box                   |
| categoryIds | array[UUID]   | Yes      | Min 1 category | Product category IDs                                     |
| description | string        | No       | Max 2000       | Product description                                      |
| attributes  | object        | No       | Free-form JSON | Dynamic attributes e.g. { color: 'red', weight: '500g' } |
| imageUrls   | array[string] | No       | Max 5 URLs     | Product image URLs (upload separately)                   |
| isActive    | boolean       | No       | -              | Visibility (default: true)                               |

**Example:**

```
{  "sku": "FRU-APPLE-001",  "name": "Organic Gala Apple",  "price": 45000,  "unit": "Kg",  "categoryIds": ["uuid-cat-1"],  "isActive": true }
```

**Response**

| Field     | Type     | Description                      |
|-----------|----------|----------------------------------|
| id        | UUID     | New product ID                   |
| sku       | string   | Product SKU                      |
| name      | string   | Product name                     |
| slug      | string   | Auto-generated URL-friendly slug |
| createdAt | datetime | Creation timestamp               |

**Error Responses**

| HTTP Code | Error Code    | Description                            |
|-----------|---------------|----------------------------------------|
| 409       | SKU_CONFLICT  | A product with this SKU already exists |
| 422       | INVALID_PRICE | Price must be greater than 0           |

 **/api/v1/products/{id}**

**Get Product Detail**

Retrieve full details of a single product by ID.

 **Auth:** Bearer token — All roles

### Path Parameters

| Parameter | Type | Required | Description |
|-----------|------|----------|-------------|
| id        | UUID | Yes      | Product ID  |

### Response

| Field       | Type     | Description              |
|-------------|----------|--------------------------|
| id          | UUID     | Product ID               |
| sku         | string   | SKU / Barcode            |
| name        | string   | Product name             |
| slug        | string   | URL slug                 |
| price       | decimal  | Retail price             |
| unit        | string   | Unit of measure          |
| description | string   | Description              |
| attributes  | object   | Dynamic attributes JSON  |
| categories  | array    | List of category objects |
| imageUrls   | array    | Image URLs               |
| isActive    | boolean  | Active status            |
| createdAt   | datetime | Creation time            |
| updatedAt   | datetime | Last update time         |

**PUT** `/api/v1/products/{id}`

### Update Product

Update product information. Price changes are logged in Price History. Publishes ProductUpdated event to Kafka.

 **Auth:** Bearer token — ROLE\_MANAGER

### Path Parameters

| Parameter | Type | Required | Description |
|-----------|------|----------|-------------|
| id        | UUID | Yes      | Product ID  |

### Request Body

| Field | Type    | Required | Constraints | Description          |
|-------|---------|----------|-------------|----------------------|
| name  | string  | No       | Max 255     | New product name     |
| price | decimal | No       | > 0         | Updated retail price |
| unit  | string  | No       | -           | Unit of measure      |

| Field       | Type          | Required | Constraints | Description                |
|-------------|---------------|----------|-------------|----------------------------|
| categoryIds | array[UUID]   | No       | -           | Updated category list      |
| description | string        | No       | -           | Updated description        |
| attributes  | object        | No       | -           | Updated dynamic attributes |
| imageUrls   | array[string] | No       | Max 5       | Updated image URLs         |
| isActive    | boolean       | No       | -           | Show/hide from storefront  |

## Error Responses

| HTTP Code | Error Code    | Description                                                 |
|-----------|---------------|-------------------------------------------------------------|
| 400       | SKU_IMMUTABLE | SKU cannot be changed after transactions have been recorded |

**DELETE** `/api/v1/products/{id}`

## Discontinue Product (Soft Delete)

Set product `isActive` = false (soft delete). Product history is preserved. Product will not appear in POS search.

 **Auth:** Bearer token — ROLE\_MANAGER

## Path Parameters

| Parameter | Type | Required | Description |
|-----------|------|----------|-------------|
| id        | UUID | Yes      | Product ID  |

## Response

Example:

```
{ "message": "Product discontinued successfully", "productId": "uuid..." }
```

## Error Responses

| HTTP Code | Error Code    | Description                                      |
|-----------|---------------|--------------------------------------------------|
| 409       | ACTIVE_ORDERS | Cannot discontinue a product with pending orders |

**GET** `/api/v1/categories`

## Get All Categories

Retrieve hierarchical category tree for product classification.

 **Auth:** Bearer token — All roles

## Response

| Field   | Type   | Description   |
|---------|--------|---------------|
| [].id   | UUID   | Category ID   |
| [].name | string | Category name |

| Field       | Type      | Description                      |
|-------------|-----------|----------------------------------|
| [].slug     | string    | URL slug                         |
| [].parentId | UUID null | Parent category ID (null = root) |
| [].imageUrl | string    | Category icon URL                |
| [].children | array     | Nested sub-categories            |

**POST** /api/v1/categories**Create Category**

Add a new product category to the system.

 **Auth:** Bearer token — ROLE\_MANAGER

**Request Body**

| Field    | Type   | Required | Constraints            | Description                        |
|----------|--------|----------|------------------------|------------------------------------|
| name     | string | Yes      | Max 255                | Category display name              |
| parentId | UUID   | No       | Must exist if provided | Parent category ID (omit for root) |
| imageUrl | string | No       | Valid URL              | Category icon/image URL            |

**GET** /api/v1/products/export

Export product list (CSV/XLSX)

**POST** /api/v1/import

Import product list (CSV/XLSX)

## 4. Inventory Service

### 4.1 Inventory & Supplier Management APIs

Tracks real-time stock levels, manages supplier relationships, and processes stock import receipts (Goods Received Notes). Consumes OrderCreated events from Kafka to auto-deduct inventory.

**GET** `/api/v1/inventory/stock`

#### Get Current Stock Levels

Retrieve current inventory stock for all products in a warehouse.

 **Auth:** Bearer token — ROLE\_MANAGER | ROLE\_CASHIER

#### Query Parameters

| Parameter   | Type    | Required | Default       | Description                                   |
|-------------|---------|----------|---------------|-----------------------------------------------|
| warehouseId | integer | No       | default store | Filter by warehouse                           |
| sku         | string  | No       | -             | Filter by specific product SKU                |
| lowStock    | boolean | No       | false         | Show only low-stock items (below alert level) |
| page        | integer | No       | 1             | Page number                                   |
| size        | integer | No       | 50            | Records per page                              |

#### Response

| Field                       | Type    | Description                           |
|-----------------------------|---------|---------------------------------------|
| content[].productSku        | string  | Product SKU                           |
| content[].productName       | string  | Product name (from Product Service)   |
| content[].warehouseId       | integer | Warehouse ID                          |
| content[].quantityOnHand    | integer | Actual available stock                |
| content[].reservedQuantity  | integer | Quantity reserved for pending orders  |
| content[].availableQuantity | integer | quantityOnHand - reservedQuantity     |
| content[].isLowStock        | boolean | True if below minimum alert threshold |

**GET** `/api/v1/inventory/stock/{sku}`

#### Get Stock for Specific Product

Check real-time stock level for a single product by SKU. Used by POS before adding to cart.

 **Auth:** Bearer token — All roles

#### Path Parameters

| Parameter | Type   | Required | Description           |
|-----------|--------|----------|-----------------------|
| sku       | string | Yes      | Product SKU / Barcode |

## Response

| Field             | Type    | Description                  |
|-------------------|---------|------------------------------|
| productSku        | string  | Product SKU                  |
| quantityOnHand    | integer | Physical stock quantity      |
| reservedQuantity  | integer | Reserved quantity            |
| availableQuantity | integer | Net available quantity       |
| isLowStock        | boolean | Below minimum threshold flag |

## GET /api/v1/suppliers

### Get All Suppliers

Retrieve paginated list of all supply partners.

 **Auth:** Bearer token — ROLE\_MANAGER

### Query Parameters

| Parameter | Type    | Required | Default | Description                              |
|-----------|---------|----------|---------|------------------------------------------|
| status    | string  | No       | ACTIVE  | Filter: ACTIVE   SUSPENDED   BLACKLISTED |
| search    | string  | No       | -       | Search by supplier name                  |
| page      | integer | No       | 1       | Page number                              |

## Response

| Field                 | Type    | Description                              |
|-----------------------|---------|------------------------------------------|
| content[].id          | integer | Supplier ID                              |
| content[].name        | string  | Company name                             |
| content[].contactInfo | object  | { phone, email, contactPerson, address } |
| content[].status      | string  | ACTIVE   SUSPENDED   BLACKLISTED         |

## POST /api/v1/suppliers

### Create Supplier

Register a new supply partner in the system.

 **Auth:** Bearer token — ROLE\_MANAGER

### Request Body

| Field             | Type   | Required | Constraints | Description           |
|-------------------|--------|----------|-------------|-----------------------|
| name              | string | Yes      | Max 255     | Supplier company name |
| contactInfo.phone | string | No       | Max 20      | Contact phone number  |

| Field                     | Type   | Required | Constraints | Description          |
|---------------------------|--------|----------|-------------|----------------------|
| contactInfo.email         | string | No       | Valid email | Contact email        |
| contactInfo.contactPerson | string | No       | Max 100     | Primary contact name |
| contactInfo.address       | string | No       | -           | Business address     |

## GET /api/v1/import-receipts

### Get Import History

Retrieve list of all stock import receipts (Goods Received Notes).

 **Auth:** Bearer token — ROLE\_MANAGER

#### Query Parameters

| Parameter  | Type    | Required | Default | Description                    |
|------------|---------|----------|---------|--------------------------------|
| supplierId | integer | No       | -       | Filter by supplier             |
| status     | string  | No       | -       | DRAFT   COMPLETED   CANCELLED  |
| from       | date    | No       | -       | Start date filter (YYYY-MM-DD) |
| to         | date    | No       | -       | End date filter                |
| page       | integer | No       | 1       | Page number                    |

#### Response

| Field                 | Type     | Description             |
|-----------------------|----------|-------------------------|
| content[].id          | integer  | Receipt ID              |
| content[].supplier    | object   | { id, name }            |
| content[].totalAmount | decimal  | Total purchase value    |
| content[].status      | string   | Receipt status          |
| content[].createdAt   | datetime | Receipt creation time   |
| content[].itemCount   | integer  | Number of product lines |

## POST /api/v1/import-receipts

### Create Import Receipt (Goods Received Note)

Record incoming stock from a supplier. Creates inventory transactions and updates stock quantities.

 **Auth:** Bearer token — ROLE\_MANAGER

#### Request Body

| Field      | Type    | Required | Constraints | Description                  |
|------------|---------|----------|-------------|------------------------------|
| supplierId | integer | Yes      | Must exist  | Supplier providing the goods |

| Field               | Type    | Required | Constraints                   | Description                         |
|---------------------|---------|----------|-------------------------------|-------------------------------------|
| warehouseId         | integer | Yes      | Must exist                    | Destination warehouse               |
| items               | array   | Yes      | Min 1 item                    | List of products being imported     |
| items[].productSku  | string  | Yes      | Must exist in Product Service | Product SKU                         |
| items[].quantity    | integer | Yes      | > 0                           | Quantity received                   |
| items[].importPrice | decimal | Yes      | > 0                           | Purchase cost price per unit        |
| items[].expiryDate  | date    | No       | Future date                   | Product expiry date (if applicable) |
| items[].batchNumber | string  | No       | Max 50                        | Batch/lot number                    |

**Example:**

```
{  "supplierId": 3,  "warehouseId": 1,  "items": [    { "productSku": "FRU-APPLE-001",    "quantity": 500, "importPrice": 30000 }  ] }
```

**Response**

| Field       | Type     | Description                       |
|-------------|----------|-----------------------------------|
| id          | integer  | New receipt ID                    |
| status      | string   | COMPLETED                         |
| totalAmount | decimal  | Total value of import             |
| itemCount   | integer  | Number of product lines processed |
| createdAt   | datetime | Receipt timestamp                 |

**Error Responses**

| HTTP Code | Error Code        | Description                                              |
|-----------|-------------------|----------------------------------------------------------|
| 422       | PRODUCT_NOT_FOUND | One or more product SKUs do not exist in Product Service |

**POST /api/v1/warehouses**

Create warehouse

**PUT /api/v1/warehouses**

Update warehouse

**GET PRODUCT BY WAREHOUSE /api/v1/{warehouse-id}/{product-id}**

Get product by warehouse

# 5. Customer & Loyalty Service

## 5.1 Customer Relationship & Loyalty Program APIs

Manages customer profiles, loyalty points, membership tiers, and point transaction history. Consumes OrderCreated events to automatically award points after successful purchases.

**GET** /api/v1/customers

### Get All Customers

Retrieve paginated list of all registered loyalty members.

 **Auth:** Bearer token — ROLE\_MANAGER

#### Query Parameters

| Parameter | Type    | Required | Default | Description                                    |
|-----------|---------|----------|---------|------------------------------------------------|
| search    | string  | No       | -       | Search by name or phone number (partial match) |
| tierLevel | string  | No       | -       | Filter by tier: SILVER   GOLD   DIAMOND        |
| page      | integer | No       | 1       | Page number                                    |
| size      | integer | No       | 20      | Records per page                               |

#### Response

| Field                   | Type     | Description                      |
|-------------------------|----------|----------------------------------|
| content[].id            | UUID     | Customer ID                      |
| content[].phone         | string   | Phone number (unique identifier) |
| content[].fullName      | string   | Customer full name               |
| content[].currentPoints | integer  | Current spendable loyalty points |
| content[].tierLevel     | string   | SILVER   GOLD   DIAMOND          |
| content[].totalSpent    | decimal  | Lifetime total spending (VND)    |
| content[].createdAt     | datetime | Registration date                |

**GET** /api/v1/customers/lookup

### Lookup Customer by Phone (POS)

Fast lookup of customer by phone number for POS checkout flow.

 **Auth:** Bearer token — ROLE\_CASHIER | ROLE\_MANAGER

#### Query Parameters

| Parameter | Type   | Required | Default | Description                         |
|-----------|--------|----------|---------|-------------------------------------|
| phone     | string | Yes      | -       | Customer phone number (exact match) |

## Response

| Field         | Type    | Description                                                   |
|---------------|---------|---------------------------------------------------------------|
| id            | UUID    | Customer ID                                                   |
| fullName      | string  | Customer name                                                 |
| phone         | string  | Phone number                                                  |
| currentPoints | integer | Available loyalty points                                      |
| tierLevel     | string  | Membership tier                                               |
| pointValue    | decimal | Monetary value of current points based on active loyalty rule |

## Error Responses

| HTTP Code | Error Code         | Description                                   |
|-----------|--------------------|-----------------------------------------------|
| 404       | CUSTOMER_NOT_FOUND | No customer registered with this phone number |

### POST /api/v1/customers

#### Register New Customer (Quick Registration)

Create a new loyalty member account at POS. Must complete in under 10 seconds. Initializes points at 0.

 **Auth:** Bearer token — ROLE\_CASHIER | ROLE\_MANAGER

#### Request Body

| Field    | Type   | Required | Constraints           | Description                           |
|----------|--------|----------|-----------------------|---------------------------------------|
| phone    | string | Yes      | Unique, Max 15 digits | Primary identifier for customer       |
| fullName | string | Yes      | Max 100               | Customer full name                    |
| dob      | date   | No       | YYYY-MM-DD            | Date of birth for birthday promotions |

#### Example:

```
{  "phone": "0912345678",  "fullName": "Tran Thi Mai" }
```

## Response

| Field         | Type    | Description                   |
|---------------|---------|-------------------------------|
| id            | UUID    | New customer ID               |
| phone         | string  | Registered phone              |
| fullName      | string  | Name                          |
| currentPoints | integer | Always 0 on registration      |
| tierLevel     | string  | Always SILVER on registration |

| Field     | Type     | Description            |
|-----------|----------|------------------------|
| createdAt | datetime | Registration timestamp |

Error Responses

| HTTP Code | Error Code     | Description                                      |
|-----------|----------------|--------------------------------------------------|
| 409       | PHONE_CONFLICT | A customer with this phone number already exists |

**GET** /api/v1/customers/{id}

Get Customer Profile

Retrieve detailed customer profile including loyalty status and summary metrics.

 **Auth:** Bearer token — ROLE\_MANAGER

Path Parameters

| Parameter | Type | Required | Description |
|-----------|------|----------|-------------|
| id        | UUID | Yes      | Customer ID |

Response

| Field         | Type     | Description          |
|---------------|----------|----------------------|
| id            | UUID     | Customer ID          |
| phone         | string   | Phone number         |
| fullName      | string   | Full name            |
| dob           | date     | Date of birth        |
| currentPoints | integer  | Spendable points     |
| tierLevel     | string   | Membership tier      |
| totalSpent    | decimal  | Total lifetime spend |
| createdAt     | datetime | Registration date    |

**GET** /api/v1/customers/{id}/point-transactions

Get Customer Point Transaction History

Retrieve full log of loyalty point transactions (earned, redeemed, expired, adjusted) for a customer.

 **Auth:** Bearer token — ROLE\_MANAGER

Path Parameters

| Parameter | Type | Required | Description |
|-----------|------|----------|-------------|
| id        | UUID | Yes      | Customer ID |

Query Parameters

| Parameter | Type    | Required | Default | Description                     |
|-----------|---------|----------|---------|---------------------------------|
| type      | string  | No       | -       | EARN   REDEEM   EXPIRE   ADJUST |
| from      | date    | No       | -       | Start date filter               |
| to        | date    | No       | -       | End date filter                 |
| page      | integer | No       | 1       | Page number                     |

## Response

| Field                     | Type     | Description                                                     |
|---------------------------|----------|-----------------------------------------------------------------|
| content[].id              | integer  | Transaction ID                                                  |
| content[].pointsAmount    | integer  | Points changed (positive = earned, negative = redeemed/expired) |
| content[].type            | string   | EARN   REDEEM   EXPIRE   ADJUST                                 |
| content[].orderId         | string   | Related order ID (if applicable)                                |
| content[].description     | string   | Human-readable reason                                           |
| content[].transactionDate | datetime | When the transaction occurred                                   |

**POST** /api/v1/customers/{id}/adjust-points

## Manual Loyalty Points Adjustment

Store Manager manually adjusts a customer's loyalty points for corrections or compensation. Action is logged in audit trail (CDC).

 **Auth:** Bearer token — ROLE\_MANAGER

## Path Parameters

| Parameter | Type | Required | Description |
|-----------|------|----------|-------------|
| id        | UUID | Yes      | Customer ID |

## Request Body

| Field        | Type    | Required | Constraints                    | Description                         |
|--------------|---------|----------|--------------------------------|-------------------------------------|
| pointsAmount | integer | Yes      | Non-zero, positive or negative | Points to add (+) or deduct (-)     |
| reason       | string  | Yes      | Max 500                        | Mandatory reason for the adjustment |

## Response

| Field      | Type | Description |
|------------|------|-------------|
| customerId | UUID | Customer ID |

| Field          | Type    | Description                      |
|----------------|---------|----------------------------------|
| previousPoints | integer | Points balance before adjustment |
| adjustment     | integer | Points adjusted                  |
| newPoints      | integer | Points balance after adjustment  |

## GET /api/v1/loyalty-rules

### Get Active Loyalty Rules

Retrieve current point earning and redemption configuration.

 **Auth:** Bearer token — All roles

#### Response

| Field                  | Type      | Description                                                            |
|------------------------|-----------|------------------------------------------------------------------------|
| [].id                  | integer   | Rule ID                                                                |
| [].name                | string    | Rule name                                                              |
| [].pointConversionRate | decimal   | Amount in VND needed to earn 1 point (e.g., 1000 = 1000 VND → 1 point) |
| [].minOrderValue       | decimal   | Minimum order value to earn points                                     |
| [].isActive            | boolean   | Whether this rule is currently active                                  |
| [].startDate           | date      | Rule effective start date                                              |
| [].endDate             | date null | Rule expiry date (null = no expiry)                                    |

## POST /api/v1/loyalty-rules

### Create / Update Loyalty Rule

Admin creates a new loyalty rule. Only one rule can be active at a time.

 **Auth:** Bearer token — ROLE\_ADMIN

#### Request Body

| Field               | Type    | Required | Constraints     | Description                                                   |
|---------------------|---------|----------|-----------------|---------------------------------------------------------------|
| name                | string  | Yes      | Max 100         | Rule label                                                    |
| pointConversionRate | decimal | Yes      | > 0             | VND per point earned (e.g., 1000)                             |
| redemptionRate      | decimal | Yes      | > 0             | VND discount per point redeemed (e.g., 100 points = 1000 VND) |
| minOrderValue       | decimal | No       | >= 0            | Minimum order value to qualify for point earning              |
| startDate           | date    | Yes      | Today or future | When this rule becomes effective                              |

| Field   | Type | Required | Constraints     | Description                                  |
|---------|------|----------|-----------------|----------------------------------------------|
| endDate | date | No       | After startDate | When this rule expires (omit for indefinite) |

## 6. Order Service

### 6.1 POS Transaction & Order Management APIs

Core transaction engine for the POS system. Handles cart checkout, payment processing, and order lifecycle management. Publishes OrderCreated events to trigger downstream inventory and loyalty updates.

**POST** `/api/v1/orders`

#### Create Order (Process Checkout)

The core POS transaction endpoint. Validates cart, applies discounts, creates the order, processes payment, and publishes OrderCreated event to Kafka (triggers inventory deduction and loyalty point update).

 **Auth:** Bearer token — ROLE\_CASHIER

#### Request Body

| Field                   | Type      | Required | Constraints                           | Description                                  |
|-------------------------|-----------|----------|---------------------------------------|----------------------------------------------|
| customerId              | UUID null | No       | -                                     | Linked customer ID (null for walk-in guests) |
| cashierId               | UUID      | Yes      | Must be logged-in cashier             | Cashier processing the transaction           |
| items                   | array     | Yes      | Min 1 item                            | Products in the cart                         |
| items[].productSku      | string    | Yes      | Must exist and be in stock            | Product SKU                                  |
| items[].quantity        | integer   | Yes      | > 0                                   | Quantity purchased                           |
| items[].unitPrice       | decimal   | Yes      | > 0                                   | Price at time of sale (snapshot)             |
| usePoints               | boolean   | No       | -                                     | Whether to apply loyalty points discount     |
| pointsToRedeem          | integer   | No       | >= 0, <= customer balance             | Number of points to redeem for discount      |
| payment.method          | string    | Yes      | CASH   CARD   QR_CODE   BANK_TRANSFER | Payment method                               |
| payment.amountReceived  | decimal   | Yes      | >= finalAmount                        | Cash received from customer                  |
| payment.transactionCode | string    | No       | Required for CARD/QR                  | Bank/payment terminal reference code         |

#### Example:

```
{
  "customerId": "uuid-cust-123",
  "cashierId": "uuid-cashier-456",
  "items": [
    { "productSku": "FRU-APPLE-001", "quantity": 2, "unitPrice": 45000 }
  ],
  "usePoints": true,
  "pointsToRedeem": 100,
  "payment": { "method": "CASH",
    "amountReceived": 200000 }
}
```

#### Response

| Field               | Type     | Description                                    |
|---------------------|----------|------------------------------------------------|
| id                  | UUID     | Order ID                                       |
| orderCode           | string   | Human-readable order code (e.g., ORD-2026-001) |
| totalAmount         | decimal  | Subtotal before discounts                      |
| discountAmount      | decimal  | Voucher/coupon discounts                       |
| pointDiscountAmount | decimal  | Discount from point redemption                 |
| finalAmount         | decimal  | Customer pays this amount                      |
| changeAmount        | decimal  | Change to return (for cash payments)           |
| pointsEarned        | integer  | Loyalty points awarded for this transaction    |
| pointsRedeemed      | integer  | Points consumed                                |
| status              | string   | COMPLETED                                      |
| createdAt           | datetime | Transaction timestamp                          |

### Error Responses

| HTTP Code | Error Code           | Description                                    |
|-----------|----------------------|------------------------------------------------|
| 400       | INSUFFICIENT_STOCK   | One or more items do not have enough stock     |
| 400       | INSUFFICIENT_PAYMENT | amountReceived is less than finalAmount        |
| 400       | INSUFFICIENT_POINTS  | Customer does not have enough points to redeem |
| 422       | PRODUCT_INACTIVE     | One or more products are discontinued          |

**GET** /api/v1/orders

### Get Order List

Retrieve paginated order history. Manager sees all orders; Cashier sees orders from their shifts.

 **Auth:** Bearer token — ROLE\_MANAGER | ROLE\_CASHIER

### Query Parameters

| Parameter  | Type    | Required | Default | Description                                |
|------------|---------|----------|---------|--------------------------------------------|
| cashierId  | UUID    | No       | -       | Filter by cashier                          |
| customerId | UUID    | No       | -       | Filter by customer                         |
| status     | string  | No       | -       | PENDING   COMPLETED   CANCELLED   RETURNED |
| from       | date    | No       | -       | Start date filter                          |
| to         | date    | No       | -       | End date filter                            |
| page       | integer | No       | 1       | Page number                                |
| size       | integer | No       | 20      | Records per page                           |

## Response

| Field                 | Type     | Description                     |
|-----------------------|----------|---------------------------------|
| content[].id          | UUID     | Order ID                        |
| content[].orderCode   | string   | Order reference code            |
| content[].customer    | object   | { id, fullName, phone } or null |
| content[].cashier     | object   | { id, fullName }                |
| content[].finalAmount | decimal  | Total paid amount               |
| content[].status      | string   | Order status                    |
| content[].createdAt   | datetime | Order time                      |

**GET** /api/v1/orders/{id}

### Get Order Detail

Retrieve full details of a single order including all items and payment info.

 **Auth:** Bearer token — ROLE\_MANAGER | ROLE\_CASHIER

### Path Parameters

| Parameter | Type | Required | Description |
|-----------|------|----------|-------------|
| id        | UUID | Yes      | Order ID    |

## Response

| Field          | Type     | Description                                                  |
|----------------|----------|--------------------------------------------------------------|
| id             | UUID     | Order ID                                                     |
| orderCode      | string   | Order reference code                                         |
| items          | array    | Line items with snapshot name, quantity, unitPrice, subTotal |
| payment        | object   | Payment details: method, amount, transactionCode             |
| totalAmount    | decimal  | Gross total                                                  |
| finalAmount    | decimal  | Net amount paid                                              |
| pointsEarned   | integer  | Points awarded                                               |
| pointsRedeemed | integer  | Points consumed                                              |
| status         | string   | Order status                                                 |
| note           | string   | Order note if any                                            |
| createdAt      | datetime | Created time                                                 |

**PATCH** /api/v1/orders/{id}/cancel

## Cancel Order

Cancel a completed or pending order. Requires a reason. Triggers inventory restoration and points reversal. Action is logged for audit.

 **Auth:** Bearer token — ROLE\_MANAGER

### Path Parameters

| Parameter | Type | Required | Description |
|-----------|------|----------|-------------|
| id        | UUID | Yes      | Order ID    |

### Request Body

| Field  | Type   | Required | Constraints | Description                                         |
|--------|--------|----------|-------------|-----------------------------------------------------|
| reason | string | Yes      | Max 500     | Mandatory cancellation reason (business rule BR-09) |

### Error Responses

| HTTP Code | Error Code            | Description                                                          |
|-----------|-----------------------|----------------------------------------------------------------------|
| 400       | ALREADY_CANCELLED     | Order is already in CANCELLED status                                 |
| 400       | RETURN_PERIOD_EXPIRED | Order is too old to be cancelled (beyond configurable return window) |

## 7. Reporting Service

### 7.1 Analytics & Report Generation APIs

Provides aggregated business analytics from the OLAP data warehouse. Revenue reports, inventory analytics, and file export capabilities. Runs independently to avoid impacting POS performance.

**GET** `/api/v1/reports/revenue`

#### Get Revenue Report

Aggregate revenue data from the OLAP data warehouse. Returns charts data for total revenue, net income, average order value. Does NOT query the operational Order Service directly to avoid performance impact.

 **Auth:** Bearer token — ROLE\_MANAGER

#### Query Parameters

| Parameter  | Type   | Required | Default | Description                        |
|------------|--------|----------|---------|------------------------------------|
| period     | string | Yes      | -       | DAILY   WEEKLY   MONTHLY   YEARLY  |
| from       | date   | Yes      | -       | Start date (YYYY-MM-DD)            |
| to         | date   | Yes      | -       | End date                           |
| categoryId | UUID   | No       | -       | Filter revenue by product category |

#### Response

| Field                 | Type    | Description                                                    |
|-----------------------|---------|----------------------------------------------------------------|
| summary.totalRevenue  | decimal | Total revenue in period (VND)                                  |
| summary.totalOrders   | integer | Total completed orders                                         |
| summary.avgOrderValue | decimal | Average order value                                            |
| summary.totalTax      | decimal | Total VAT collected                                            |
| summary.netRevenue    | decimal | Revenue after tax                                              |
| chartData             | array   | Array of { date, revenue, orderCount } for chart rendering     |
| topProducts           | array   | Top 10 best-selling products: { sku, name, quantity, revenue } |

**GET** `/api/v1/reports/inventory`

#### Get Inventory Report

Provide inventory analytics: best-selling products, slow-moving stock, and low-stock alerts.

 **Auth:** Bearer token — ROLE\_MANAGER

#### Query Parameters

| Parameter | Type | Required | Default | Description         |
|-----------|------|----------|---------|---------------------|
| from      | date | Yes      | -       | Report period start |

| Parameter    | Type    | Required | Default | Description                 |
|--------------|---------|----------|---------|-----------------------------|
| to           | date    | Yes      | -       | Report period end           |
| lowStockOnly | boolean | No       | false   | Return only low-stock items |

## Response

| Field               | Type    | Description                                                               |
|---------------------|---------|---------------------------------------------------------------------------|
| lowStockItems       | array   | Products below minimum threshold: { sku, name, quantityOnHand, minLevel } |
| bestSellers         | array   | Top selling products in period: { sku, name, totalSold, revenue }         |
| slowMoving          | array   | Products with low turnover rate                                           |
| totalInventoryValue | decimal | Estimated total value of current stock                                    |

**POST** /api/v1/reports/export

## Export Report to File

Trigger generation of a PDF or XLSX report file. Returns a download URL.

 **Auth:** Bearer token — ROLE\_MANAGER

## Request Body

| Field      | Type   | Required | Constraints                   | Description                |
|------------|--------|----------|-------------------------------|----------------------------|
| reportType | string | Yes      | REVENUE   INVENTORY   PAYROLL | Type of report to generate |
| format     | string | Yes      | PDF   XLSX                    | Output file format         |
| from       | date   | Yes      | YYYY-MM-DD                    | Period start date          |
| to         | date   | Yes      | YYYY-MM-DD                    | Period end date            |

## Response

| Field       | Type     | Description                                                         |
|-------------|----------|---------------------------------------------------------------------|
| downloadUrl | string   | Signed URL to download the generated report (expires in 15 minutes) |
| fileName    | string   | Report file name                                                    |
| generatedAt | datetime | Generation timestamp                                                |

# 8. Notification Service

## 8.1 Notification & Messaging APIs

Handles delivery of system messages via email, SMS, and push notifications. Triggered by system events (OTP requests, low stock alerts, payslip delivery). Supports template-based messaging.

**POST** /api/v1/notifications/send

### Send Notification (Internal)

Internal endpoint used by other services to queue a notification (email, SMS, push). Not directly called by frontend — triggered by system events.

 **Auth:** Internal service-to-service token

#### Request Body

| Field        | Type   | Required | Constraints                          | Description                                                            |
|--------------|--------|----------|--------------------------------------|------------------------------------------------------------------------|
| recipientId  | string | Yes      | -                                    | Recipient identifier (email, phone, or user ID)                        |
| type         | string | Yes      | EMAIL   SMS   PUSH_APP               | Notification channel                                                   |
| templateCode | string | Yes      | Must exist in notification_templates | e.g., OTP_SMS, ORDER_SUCCESS_EMAIL, LOW_STOCK_ALERT                    |
| content      | object | Yes      | -                                    | Dynamic data to fill template placeholders, e.g. { name, amount, otp } |

#### Response

| Field          | Type   | Description                  |
|----------------|--------|------------------------------|
| notificationId | UUID   | Queued notification ID       |
| status         | string | PENDING (queued for sending) |

**GET** /api/v1/notifications

### Get Notification History

Retrieve log of sent notifications for auditing and debugging.

 **Auth:** Bearer token — ROLE\_ADMIN

#### Query Parameters

| Parameter | Type    | Required | Default | Description             |
|-----------|---------|----------|---------|-------------------------|
| status    | string  | No       | -       | PENDING   SENT   FAILED |
| type      | string  | No       | -       | EMAIL   SMS   PUSH_APP  |
| from      | date    | No       | -       | Start date              |
| page      | integer | No       | 1       | Page                    |

## Response

| Field                  | Type     | Description      |
|------------------------|----------|------------------|
| content[].id           | UUID     | Notification ID  |
| content[].recipientId  | string   | Recipient        |
| content[].type         | string   | Channel          |
| content[].templateCode | string   | Template used    |
| content[].status       | string   | Delivery status  |
| content[].retryCount   | integer  | Retry attempts   |
| content[].sentAt       | datetime | Actual send time |

**GET** /api/v1/notification-templates

## Get Notification Templates

Retrieve all notification templates configured in the system.

 **Auth:** Bearer token — ROLE\_ADMIN

## Response

| Field           | Type    | Description                              |
|-----------------|---------|------------------------------------------|
| [].id           | UUID    | Template ID                              |
| [].code         | string  | Unique template code                     |
| [].title        | string  | Subject or push title                    |
| [].bodyTemplate | string  | HTML/text template with {{placeholders}} |
| [].channel      | string  | EMAIL   SMS   PUSH                       |
| [].isActive     | boolean | Active status                            |

## 9. System Configuration & Audit

### 9.1 Store Configuration & Audit Log APIs

Admin-only APIs for configuring store information, VAT/tax rules, and viewing the immutable system audit trail (CDC logs). All sensitive actions are automatically tracked.

**GET** `/api/v1/system/config`

#### Get System Configuration

Retrieve all system-wide configuration: store info, VAT settings.

 **Auth:** Bearer token — All roles (read-only for non-Admin)

#### Response

| Field                 | Type    | Description                                       |
|-----------------------|---------|---------------------------------------------------|
| store.name            | string  | Store name                                        |
| store.phone           | string  | Store phone number                                |
| store.address         | string  | Store address                                     |
| store.logoUrl         | string  | Store logo URL                                    |
| receipt.footerMessage | string  | Receipt footer text                               |
| receipt.showQrCode    | boolean | Enable QR code on receipts                        |
| tax                   | array   | List of tax rules: { name, percentage, isActive } |

**PUT** `/api/v1/system/config/store`

#### Update Store Information

Admin updates the store profile. Changes are reflected on all receipts immediately.

 **Auth:** Bearer token — ROLE\_ADMIN

#### Request Body

| Field                 | Type    | Required | Constraints | Description            |
|-----------------------|---------|----------|-------------|------------------------|
| name                  | string  | Yes      | Max 100     | Store name             |
| phone                 | string  | Yes      | Max 20      | Contact phone          |
| address               | string  | Yes      | Max 200     | Physical address       |
| logoUrl               | string  | No       | Valid URL   | Logo image URL         |
| receipt.footerMessage | string  | No       | Max 300     | Receipt footer message |
| receipt.showQrCode    | boolean | No       | -           | QR code on receipts    |

**PUT** `/api/v1/system/config/tax`

#### Update VAT / Tax Settings

Admin updates tax rules. Changes apply to all future transactions. Historical orders are NOT affected (Business Rule BR-03).

 **Auth:** Bearer token — ROLE\_ADMIN

### Request Body

| Field                 | Type    | Required | Constraints | Description              |
|-----------------------|---------|----------|-------------|--------------------------|
| taxRules              | array   | Yes      | -           | List of tax rules to set |
| taxRules[].name       | string  | Yes      | Max 50      | Tax name (e.g., VAT)     |
| taxRules[].percentage | decimal | Yes      | >= 0        | Tax rate percentage      |
| taxRules[].isActive   | boolean | Yes      | -           | Enable/disable this tax  |

### GET /api/v1/audit-logs

### Get System Audit Logs

Retrieve immutable system audit trail. Admin only. Logs cannot be deleted or modified by anyone.

 **Auth:** Bearer token — ROLE\_ADMIN

### Query Parameters

| Parameter | Type     | Required | Default  | Description                                                                 |
|-----------|----------|----------|----------|-----------------------------------------------------------------------------|
| type      | string   | No       | -        | CDC   ACCESS   MICROSERVICE                                                 |
| actor     | string   | No       | -        | Filter by user or service name                                              |
| action    | string   | No       | -        | Filter by action:<br>PRICE_CHANGE  <br>POINT_MOD  <br>PERMISSION_GRANT etc. |
| from      | datetime | No       | Last 24h | Start of time range                                                         |
| to        | datetime | No       | Now      | End of time range                                                           |
| page      | integer  | No       | 1        | Page number                                                                 |
| size      | integer  | No       | 50       | Records per page                                                            |

### Response

| Field                 | Type     | Description                                   |
|-----------------------|----------|-----------------------------------------------|
| content[].timestamp   | datetime | When the action occurred                      |
| content[].actor       | string   | User or service that performed the action     |
| content[].type        | string   | Log type (CDC, ACCESS, etc.)                  |
| content[].action      | string   | Action performed                              |
| content[].reason      | string   | Description / metadata                        |
| content[].isSensitive | boolean  | True for critical actions (highlighted in UI) |

| Field               | Type   | Description            |
|---------------------|--------|------------------------|
| content[].ipAddress | string | Originating IP address |

**GET** `/api/v1/audit-logs/{id}`

### Get Audit Log Detail

Retrieve full detail of a specific audit log entry including before/after state for CDC logs.

 **Auth:** Bearer token — ROLE\_ADMIN

#### Path Parameters

| Parameter | Type | Required | Description        |
|-----------|------|----------|--------------------|
| id        | UUID | Yes      | Audit log entry ID |

#### Response

| Field        | Type     | Description                                  |
|--------------|----------|----------------------------------------------|
| id           | UUID     | Log ID                                       |
| timestamp    | datetime | Event time                                   |
| actor        | string   | Performer                                    |
| action       | string   | Action type                                  |
| targetEntity | string   | Affected entity (e.g., User, Product, Order) |
| targetId     | string   | ID of affected entity                        |
| beforeState  | object   | Data snapshot before the change (CDC logs)   |
| afterState   | object   | Data snapshot after the change (CDC logs)    |